A NEW APPROACH TO EQUILIBRIUM PROBLEMS VIA REGULARIZED EXTRAGRADIENT-TYPE DYNAMICS

SUPPLEMENTARY MATERIAL: DETAILED PROOF

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A new approach to equilibrium problems via regularized extragradient-type dynamics

Let C be a nonempty, closed and convex set in a real Hilbert space H, $f : H \times H \to \mathbb{R}$ be an equilibrium bifunction, i.e., $f(x,x) = 0$ for all $x \in H$. Consider the following equilibrium problem

$$\text{Find } x^* \in C \text{ such that } f(x^*, y) \geq 0 \quad \forall y \in C. \tag{1}$$

Denote by $Sol(f,C)$ the solution set of (1). In what follows, we assume that $Sol(f,C)$ is not empty. We focus on the following dynamics associated with (1):

$$\begin{cases} z(t) = \text{prox}_{\lambda(t)(f(x(t),\cdot)+\alpha(t)g(x(t),\cdot))}(x(t)), \\ y(t) = \text{prox}_{\lambda(t)(f(x(t),\cdot)+\alpha(t)g(z(t),\cdot))}(z(t)), \\ \dot{x}(t) + x(t) = y(t), \end{cases} \tag{2}$$

or equivalently,

$$\begin{cases} z(t) = \text{argmin}\{\lambda(t)f(x(t),u) + \lambda(t)\alpha(t)g(x(t),u) + \frac{1}{2}\|u - x(t)\|^2 : u \in C\}, \\ y(t) = \text{argmin}\{\lambda(t)f(x(t),v) + \lambda(t)\alpha(t)g(z(t),v) + \frac{1}{2}\|v - z(t)\|^2 : v \in C\}, \\ \dot{x}(t) + x(t) = y(t). \end{cases} \tag{3}$$

Discretizing (2) by approximating $\dot{x}(t_k) \approx \dfrac{x(t_{k+1}) - x(t_k)}{\tau_k}$ leads to several regularized extragradient-type algorithms with relaxation parameters, namely
REA1 (Regularized Extragradient-Type Algorithm 1)

$$\begin{cases} z^k = \text{prox}_{\lambda_k\left(f(x^k,\cdot)+\alpha_k g(x^k,\cdot)\right)}(x^k), \\ y^k = \text{prox}_{\lambda_k\left(f(x^k,\cdot)+\alpha_k g(z^k,\cdot)\right)}(z^k), \\ x^{k+1} = (1-\tau_k)x^k + \tau_k y^k. \end{cases} \tag{4}$$

REA2 (Regularized Extragradient-Type Algorithm 2)

$$\begin{cases} z^k = \text{prox}_{\lambda_k\left(f(x^k,\cdot)+\alpha_k g(x^k,\cdot)\right)}(x^k), \\ y^k = \text{prox}_{\lambda_{k+1}\left(f(x^k,\cdot)+\alpha_{k+1} g(z^k,\cdot)\right)}(z^k), \\ x^{k+1} = (1-\tau_k)x^k + \tau_k y^k. \end{cases} \tag{5}$$

The following issues should be solved:

i. The existence and uniqueness of the strong global solution $x(t)$ to (2).

ii. The strong convergence of $x(t)$ to a special chosen solution $x^\dagger$ of the equilibrium problem (1).

iii. The strong convergence of regularized extragardient-type algorithms REA1, REA2 to the solution $x^\dagger$.

iv. Convergence rate of the REA1, REA2 under the strong monotonicity of the bifunction $f(x,y)$.

v. Numerical experiments.

Answer for iii. We first bring out the following hypotheses for a bifunction $f : C \times C \to \mathbb{R}$:
(H1) f is monotone and $f(u, u) = 0$ for all $u \in C$.
(H1') f is γ-strongly monotone.
(H2) $f(u, .)$ is convex and weakly lower semi-continuous, $f(., v)$ is weakly upper semi-continuous.
(H3) The solution set $Sol(f, C)$ is nonempty.
Now, we take algorithm REA1 into consideration.
Statement 1. Assume that hypotheses (H1), (H2), (H3) hold for the bifunction f; and (H1'), (H2) hold for the (uniformly bounded) bifunction g. Moreover, f satisfies the L-Lipschitz type condition and g satisfies the K-Lipschitz type condition. Besides, the sequences $\{\tau_k\} \subset [a, b] \subset (0, 1)$, $\{\lambda_k\}$, $\{\alpha_k\} \subset (0, +\infty)$ satisfy

$$\sum_{k=0}^{+\infty} \lambda_k \alpha_k = +\infty; \quad \lim_{k\to\infty} \alpha_k = 0; \quad \lim_{k\to\infty} \frac{\lambda_k}{\alpha_k} = 0; \quad \lim_{k\to\infty} \frac{|\alpha_k - \alpha_{k+1}|}{\alpha_k^2} = 0. \tag{6}$$

Then, the sequence $\{x^k\}$ generated by algorithm REA1 converges strongly to a solution $x^\dagger$ of problem 1, where $x^\dagger \in Sol(g, Sol(f, C))$.
Proof of statement 1. For each $\alpha > 0$, let x_α denote the unique solution of a regularized problem

$$\text{Find } x^* \in C \text{ such that } f(x^*, y) + \alpha g(x^*, y) \geq 0 \quad \forall y \in C. \tag{7}$$

It is sufficient to prove that $x^k - x_{\alpha_k}$ as $k \to \infty$ since we already have $x_{\alpha_k} \to x^\dagger$. We see that

$$\begin{aligned}
||x^{k+1} - x_{\alpha_{k+1}}||^2 &= ||(1-\tau_k)x^k + \tau_k y^k - x_{\alpha_{k+1}}||^2 \\
&= ||(x^k - x_{\alpha_k}) + \tau_k(y^k - x^k) + (x_{\alpha_k} - x_{\alpha_{k+1}})||^2 \\
&= ||x^k - x_{\alpha_k}||^2 + \tau_k^2||y^k - x^k||^2 + ||x_{\alpha_k} - x_{\alpha_{k+1}}||^2 \\
&+ 2\langle x^k - x_{\alpha_k}, x_{\alpha_k} - x_{\alpha_{k+1}}\rangle + 2\tau_k\langle y^k - x^k, x_{\alpha_k} - x_{\alpha_{k+1}}\rangle + 2\tau_k\langle x^k - x_{\alpha_k}, y^k - x^k\rangle \\
&\leq ||x^k - x_{\alpha_k}||^2 + \tau_k^2||x^k - y^k||^2 + M^2\frac{|\alpha_k - \alpha_{k+1}|^2}{\alpha_k^2} + 2M\frac{|\alpha_k - \alpha_{k+1}|}{\alpha_k}||x^k - x_{\alpha_k}|| \\
&+ 2\tau_k M\frac{|\alpha_k - \alpha_{k+1}|}{\alpha_k}||x^k - y^k|| + 2\tau_k\langle x^k - x_{\alpha_k}, y^k - x^k\rangle.
\end{aligned} \tag{8}$$

Next, we have

$$\begin{aligned}
\langle x^k - x_{\alpha_k}, y^k - x^k\rangle &= \langle y^k - x_{\alpha_k}, y^k - x^k\rangle - ||x^k - y^k||^2 \\
&= \langle y^k - x_{\alpha_k}, y^k - z^k\rangle + \langle z^k - x_{\alpha_k}, z^k - x^k\rangle + \langle y^k - z^k, z^k - x^k\rangle - ||x^k - y^k||^2.
\end{aligned} \tag{9}$$

Because $y^k = \text{prox}_{\lambda_k\left(f(x^k,\cdot)+\alpha_k g(z^k,\cdot)\right)}(z^k)$, it follows from lemma 1 in [1] that for all $v \in C$,

$$\langle y^k - v, y^k - z^k\rangle \leq \lambda_k[f(x^k, v) + \alpha_k g(z^k, v)] - \lambda_k[f(x^k, y^k) + \alpha_k g(z^k, y^k)].$$

We substitute $v = x_{\alpha_k}$ into this inequality and re-organize to obtain

$$\langle y^k - x_{\alpha_k}, y^k - z^k\rangle \leq \lambda_k[f(x^k, x_{\alpha_k}) - f(x^k, y^k)] + \lambda_k\alpha_k[g(z^k, x_{\alpha_k}) - g(z^k, y^k)]. \tag{10}$$

It follows from the Lipschitz type condition of f and g that

$$f(x^k, x_{\alpha_k}) - f(x^k, y^k) \leq L||x^k - y^k||||x_{\alpha_k} - y^k|| + f(y^k, x_{\alpha_k}) \tag{11}$$

and

$$g(z^k, x_{\alpha_k}) - g(z^k, y^k) \leq K||z^k - y^k||||x_{\alpha_k} - y^k|| + g(y^k, x_{\alpha_k}). \tag{12}$$

We combine (10), (11) and (12) to get

$$\begin{aligned}\langle y^k - x_{\alpha_k}, y^k - z^k\rangle \leq \lambda_k L||x^k - y^k||||x_{\alpha_k} - y^k|| + \lambda_k\alpha_k K||z^k - y^k||||x_{\alpha_k} - y^k|| \\ + \lambda_k[f(y^k, x_{\alpha_k}) + \alpha_k g(y^k, x_{\alpha_k})].\end{aligned} \tag{13}$$

Since x_{α_k} is solution of a problem (7), we have

$$f(x_{\alpha_k}, y^k) + \alpha_k g(x_{\alpha_k}, y^k) \geq 0. \tag{14}$$

The monotonicity of f together with the γ-strong monotonicity of g implies that

$$[f(x_{\alpha_k}, y^k) + f(y^k, x_{\alpha_k})] + \alpha_k[g(x_{\alpha_k}, y^k) + g(y^k, x_{\alpha_k})] \leq -\gamma\alpha_k||x_{\alpha_k} - y^k||^2,$$

or equivalently,

$$[f(x_{\alpha_k}, y^k) + \alpha_k g(x_{\alpha_k}, y^k)] + [f(y^k, x_{\alpha_k}) + \alpha_k g(y^k, x_{\alpha_k})] \leq -\gamma\alpha_k||x_{\alpha_k} - y^k||^2. \tag{15}$$

From (14) and (15), we get

$$f(y^k, x_{\alpha_k}) + \alpha_k g(y^k, x_{\alpha_k}) \leq -\gamma\alpha_k||x_{\alpha_k} - y^k||^2. \tag{16}$$

We combine (13) and (16) to reach

$$\begin{aligned}\langle y^k - x_{\alpha_k}, y^k - z^k\rangle \leq \lambda_k L||x^k - y^k||||x_{\alpha_k} - y^k|| + \lambda_k\alpha_k K||z^k - y^k||||x_{\alpha_k} - y^k|| \\ - \gamma\lambda_k\alpha_k||x_{\alpha_k} - y^k||^2.\end{aligned} \tag{17}$$

Next, because $z^k = \text{prox}_{\lambda_k\left(f(x^k,\cdot)+\alpha_k g(x^k,\cdot)\right)}(x^k)$, we once again use lemma 1 in [1] to get

$$\langle z^k - v, z^k - x^k\rangle \leq \lambda_k[f(x^k, v) + \alpha_k g(x^k, v)] - \lambda_k[f(x^k, z^k) + \alpha_k g(x^k, z^k)]$$

for all $v \in C$. We then substitute $v = x_{\alpha_k}$ and repeat the transformations from (10) to (17) to see that

$$\begin{aligned}\langle z^k - x_{\alpha_k}, z^k - x^k\rangle &\leq \lambda_k L||x^k - z^k||||x_{\alpha_k} - z^k|| + \lambda_k\alpha_k K||x^k - z^k||||x_{\alpha_k} - z^k|| \\ &\quad - \gamma\lambda_k\alpha_k||x_{\alpha_k} - z^k||^2 \\ &= \lambda_k(L + \alpha_k K)||x^k - z^k||||x_{\alpha_k} - z^k|| - \gamma\lambda_k\alpha_k||x_{\alpha_k} - z^k||^2\end{aligned} \tag{18}$$

On the other hand, it is easy to see that

$$\begin{aligned}\langle y^k - z^k, z^k - x^k\rangle &= \langle y^k - z^k, y^k - x^k\rangle - ||y^k - z^k||^2 \\ &\leq \frac{1}{2}\left(||y^k - z^k||^2 + ||y^k - x^k||^2\right) - ||y^k - z^k||^2 \\ &= \frac{1}{2}||y^k - x^k||^2 - \frac{1}{2}||y^k - z^k||^2.\end{aligned} \tag{19}$$

We infer from (9), (17), (18) and (19) that

$$\begin{aligned}\langle x^k - x_{\alpha_k}, y^k - x^k\rangle \le\ & \lambda_k L||x^k - y^k||||x_{\alpha_k} - y^k|| + \lambda_k\alpha_k K||z^k - y^k||||x_{\alpha_k} - y^k|| \\ & - \gamma\lambda_k\alpha_k||x_{\alpha_k} - y^k||^2 + \lambda_k(L + \alpha_k K)||x^k - z^k||||x_{\alpha_k} - z^k|| \\ & - \gamma\lambda_k\alpha_k||x_{\alpha_k} - z^k||^2 - \frac{1}{2}||y^k - x^k||^2 - \frac{1}{2}||y^k - z^k||^2. \end{aligned} \tag{20}$$

We combine (8) and (20) to get

$$\begin{aligned}||x^{k+1} - x_{\alpha_{k+1}}||^2 \le\ & ||x^k - x_{\alpha_k}||^2 + (\tau_k^2 - \tau_k)||x^k - y^k||^2 + \frac{M^2|\alpha_k - \alpha_{k+1}|^2}{\alpha_k^2} + 2M\frac{|\alpha_k - \alpha_{k+1}|}{\alpha_k}||x^k - x_{\alpha_k}|| \\ & + 2\tau_k M\frac{|\alpha_k - \alpha_{k+1}|}{\alpha_k}||x^k - y^k|| + 2\tau_k\lambda_k L||x^k - y^k||||x_{\alpha_k} - y^k|| \\ & + 2\tau_k\lambda_k\alpha_k K||z^k - y^k||||x_{\alpha_k} - y^k|| + 2\tau_k\lambda_k(L + \alpha_k K)||x^k - z^k||||x_{\alpha_k} - z^k|| \\ & - 2\gamma\tau_k\lambda_k\alpha_k||x_{\alpha_k} - y^k||^2 - 2\gamma\tau_k\lambda_k\alpha_k||x_{\alpha_k} - z^k||^2 - \tau_k||y^k - z^k||^2. \end{aligned} \tag{21}$$

Since $\tau_k \in [a, b] \subset (0, 1)$, we can choose a number $c > 0$ such that

$$\tau_k^2 - \tau_k \le -4c \text{ for every } k.$$

Indeed, the above inequality holds for $c = \max\left\{\frac{1}{4}(a - a^2), \frac{1}{4}(b - b^2)\right\}$. Thus, we have

$$(\tau_k^2 - \tau_k)||x^k - y^k||^2 \le -4c||x^k - y^k||^2. \tag{22}$$

Next, we also choose a positive number $c_1 > 0$ such that $c_1 \le \min\{\frac{a}{2}, c\}$. Using Cauchy-Schwartz's inequality, we see that

$$2\tau_k\lambda_k L||x^k - y^k||||x_{\alpha_k} - y^k|| \le c||x^k - y^k||^2 + \frac{\tau_k^2\lambda_k^2L^2}{c}||x_{\alpha_k} - y^k||^2, \tag{23}$$

$$2\tau_k\lambda_k\alpha_k K||z^k - y^k||||x_{\alpha_k} - y^k|| \le \frac{\tau_k}{2}||z^k - y^k||^2 + 2\tau_k\lambda_k^2\alpha_k^2K^2||x_{\alpha_k} - y^k||^2, \tag{24}$$

and

$$\begin{aligned}2\tau_k\lambda_k(L + \alpha_k K)||x^k - z^k||||x_{\alpha_k} - z^k|| & \le \frac{c_1}{2}||x^k - z^k||^2 + \frac{2\tau_k^2\lambda_k^2(L + \alpha_k K)^2}{c_1}||x_{\alpha_k} - z^k||^2 \\ & \le c_1||x^k - y^k||^2 + c_1||y^k - z^k||^2 + \frac{2\tau_k^2\lambda_k^2(L + \alpha_k K)^2}{c_1}||x_{\alpha_k} - z^k||^2 \end{aligned} \tag{25}$$

where we used the fact that $\frac{1}{2}||x^k - z^k||^2 \le ||x^k - y^k||^2 + ||y^k - z^k||^2$ in (25). For simplicity, we set

$$\begin{cases} \mathrm{SH1}_k = \left(\frac{\tau_k^2\lambda_k^2L^2}{c} + 2\tau_k\lambda_k^2\alpha_k^2K^2 - \gamma\tau_k\lambda_k\alpha_k\right)||x_{\alpha_k} - y^k||^2, \\ \mathrm{SH2}_k = \left(\frac{2\tau_k^2\lambda_k^2(L + \alpha_k K)^2}{c_1} - 2\gamma\tau_k\lambda_k\alpha_k\right)||x_{\alpha_k} - z^k||^2, \\ \mathrm{SH3}_k = \left(c_1 - \frac{\tau_k}{2}\right)||y^k - z^k||^2. \end{cases}$$

It follows from (21), (22), (23), (24) and (25) that

$$||x^{k+1}-x_{\alpha_{k+1}}||^2 \leq ||x^k-x_{\alpha_k}||^2 - 2c||x^k-y^k||^2 + \frac{M^2|\alpha_k-\alpha_{k+1}|^2}{\alpha_k^2} + 2M\frac{|\alpha_k-\alpha_{k+1}|}{\alpha_k}||x^k-x_{\alpha_k}|| \\ + 2\tau_k M\frac{|\alpha_k-\alpha_{k+1}|}{\alpha_k}||x^k-y^k|| + \mathrm{SH1}_k + \mathrm{SH2}_k + \mathrm{SH3}_k - \gamma\tau_k\lambda_k\alpha_k||x_{\alpha_k}-y^k||^2. \tag{26}$$

It is easy to see that

$$||x_{\alpha_k}-y^k||^2 \geq \frac{1}{2}||x^k-x_{\alpha_k}||^2 - ||x^k-y^k||^2. \tag{27}$$

The relation (26) together with (27) and Cauchy-Schwartz's inequality leads to

$$\begin{aligned} ||x^{k+1}-x_{\alpha_{k+1}}||^2 &\leq ||x^k-x_{\alpha_k}||^2 - 2c||x^k-y^k||^2 + \frac{M^2|\alpha_k-\alpha_{k+1}|^2}{\alpha_k^2} \\ &+ \left(\frac{\gamma\tau_k\lambda_k\alpha_k}{4}||x^k-x_{\alpha_k}||^2 + \frac{4M^2|\alpha_k-\alpha_{k+1}|^2}{\gamma\tau_k\lambda_k\alpha_k^3}\right) + \left(c||x^k-y^k||^2 + \frac{\tau_k^2M^2|\alpha_k-\alpha_{k+1}|^2}{c\alpha_k^2}\right) \\ &+ \mathrm{SH1}_k + \mathrm{SH2}_k + \mathrm{SH3}_k - \gamma\tau_k\lambda_k\alpha_k\left(\frac{1}{2}||x^k-x_{\alpha_k}||^2 - ||x^k-y^k||^2\right). \\ &= \left(1-\frac{\gamma\tau_k\lambda_k\alpha_k}{4}\right)||x^k-x_{\alpha_k}||^2 + (\gamma\tau_k\lambda_k\alpha_k - c)\,||x^k-y^k||^2 \\ &+ \left(\frac{\tau_k^2}{c}+1\right)\frac{M^2|\alpha_k-\alpha_{k+1}|^2}{\alpha_k^2} + \frac{4M^2|\alpha_k-\alpha_{k+1}|^2}{\gamma\tau_k\lambda_k\alpha_k^3} + \mathrm{SH1}_k + \mathrm{SH2}_k + \mathrm{SH3}_k. \end{aligned} \tag{28}$$

According to the conditions that $\{h_k\}$ is bounded, $\lim \alpha_k = 0$ and $\lim \frac{\lambda_k}{\alpha_k} = 0$ as $k \to \infty$, without loss of generality we may assume that

$$\begin{cases} \dfrac{\tau_k^2\lambda_k^2L^2}{c} + 2\tau_k\lambda_k^2\alpha_k^2K^2 - \gamma\tau_k\lambda_k\alpha_k \leq 0, \\ \dfrac{2\tau_k^2\lambda_k^2(L+\alpha_kK)^2}{c_1} - 2\gamma\tau_k\lambda_k\alpha_k \leq 0, \\ \gamma\tau_k\lambda_k\alpha_k - c \leq 0. \end{cases} \tag{29}$$

It follows from (29) and the definition of c_1 that $\mathrm{SH1}_k$, $\mathrm{SH2}_k$, $\mathrm{SH3}_k$ and $(\gamma\tau_k\lambda_k\alpha_k - c)\,||x^k - y^k||^2$ are all non-positive. Therefore, we imply from (28) that

$$||x^{k+1}-x_{\alpha_{k+1}}||^2 \leq \left(1-\frac{\gamma\tau_k\lambda_k\alpha_k}{4}\right)||x^k-x_{\alpha_k}||^2 + \left(\frac{\tau_k^2}{c}+1\right)\frac{M^2|\alpha_k-\alpha_{k+1}|^2}{\alpha_k^2} + \frac{4M^2|\alpha_k-\alpha_{k+1}|^2}{\gamma\tau_k\lambda_k\alpha_k^3}. \tag{30}$$

We denote

$$\psi_k = ||x^k - x_{\alpha_k}||^2;$$
$$p_k = \frac{\gamma\tau_k\lambda_k\alpha_k}{4};$$
$$q_k = \left(\frac{\tau_k^2}{c} + 1\right)\frac{M^2|\alpha_k - \alpha_{k+1}|^2}{\alpha_k^2} + \frac{4M^2|\alpha_k - \alpha_{k+1}|^2}{\gamma\tau_k\lambda_k\alpha_k^3}.$$

From (6), we have $\lambda_k\alpha_k \to 0$ as $k \to \infty$ and $\sum_{k=0}^{\infty}\lambda_k\alpha_k = +\infty$. Thus, we see that

$$\lim_{k\to\infty} p_k = 0 \quad \text{and} \quad \sum_{k=0}^{\infty} p_k = +\infty \tag{31}$$

and it can be assumed without loss of generality that

$$p_k \in (0, 1) \tag{32}$$

for all k. Furthermore, using the condition that

$$\lim_{k\to\infty} \frac{|\alpha_k - \alpha_{k+1}|}{\lambda_k\alpha_k^2} = 0,$$

we get

$$\frac{M^2|\alpha_k - \alpha_{k+1}|^2}{\alpha_k^2}.\frac{1}{p_k} = \frac{4M^2}{\gamma\tau_k}.\left(\frac{|\alpha_k - \alpha_{k+1}|}{\lambda_k\alpha_k^2}\right)^2.\lambda_k\alpha_k \to 0 \text{ as } k \to \infty, \tag{33}$$

and

$$\frac{4M^2|\alpha_k - \alpha_{k+1}|^2}{\gamma\tau_k\lambda_k\alpha_k^3}.\frac{1}{p_k} = \frac{16M^2}{\gamma^2\tau_k^2}.\left(\frac{|\alpha_k - \alpha_{k+1}|}{\lambda_k\alpha_k^2}\right)^2 \to 0 \text{ as } k \to \infty. \tag{34}$$

We combine (33), (34) and the fact that the sequence $\{\tau_k\}$ is bounded to imply that

$$\lim_{k\to\infty} \frac{q_k}{p_k} = 0. \tag{35}$$

Consequently, we have

$$\psi_{k+1} \leq (1 - p_k)\psi_k + q_k$$

with the sequences $\{p_k\}$ and $\{q_k\}$ satisfy (31), (32) and (35). Using lemma 2 from [1], we see that $\psi_k \to 0$ as $k \to \infty$ or equivalently, $||x^k - x_{\alpha_k}||$ converges strongly to 0. The proof is complete.

Statement 2. Assume that all the conditions in statement 1 holds. Then, the sequence $\{x^k\}$ generated by algorithm REA2 converges strongly to a solution $x^\dagger$ of problem 1, where $x^\dagger \in Sol(g, Sol(f, C))$.

Proof of statement 2. We repeat the same proof as statement 1 with slight changes. Notice that since $y^k = \mathrm{prox}_{\lambda_{k+1}\left(f(x^k,\cdot)+\alpha_{k+1}g(z^k,\cdot)\right)}(z^k)$, we have

$$\begin{aligned}\langle y^k - x_{\alpha_k}, y^k - z^k\rangle \le{}& \lambda_{k+1}L||x^k - y^k||||x_{\alpha_k} - y^k|| + \lambda_{k+1}\alpha_{k+1}K||z^k - y^k||||x_{\alpha_{k+1}} - y^k|| \\ &- \gamma\lambda_{k+1}\alpha_{k+1}||x_{\alpha_k} - y^k||^2. \end{aligned} \tag{36}$$

Using (36) and the same arguments from the proof of statement 1, we finally get

$$||x^{k+1} - x_{\alpha_{k+1}}||^2 \le \left(1 - \frac{\gamma\tau_k\lambda_{k+1}\alpha_{k+1}}{4}\right)||x^k - x_{\alpha_k}||^2 + \left(\frac{\tau_k^2}{c} + 1\right)\frac{M^2|\alpha_k - \alpha_{k+1}|^2}{\alpha_k^2} + \frac{4M^2|\alpha_k - \alpha_{k+1}|^2}{\gamma\tau_k\lambda_{k+1}\alpha_{k+1}^3}. \tag{37}$$

We denote

$$\begin{aligned}\psi_k &= ||x^k - x_{\alpha_k}||^2;\\ p_k &= \frac{\gamma\tau_k\lambda_{k+1}\alpha_{k+1}}{4};\\ q_k &= \left(\frac{\tau_k^2}{c} + 1\right)\frac{M^2|\alpha_k - \alpha_{k+1}|^2}{\alpha_k^2} + \frac{4M^2|\alpha_k - \alpha_{k+1}|^2}{\gamma\tau_k\lambda_{k+1}\alpha_{k+1}^3}.\end{aligned}$$

Therefore, the relation (37) can be rewritten as

$$\psi_{k+1} \le (1 - p_k)\psi_k + q_k. \tag{38}$$

It is easy to see that $\lim_{k\to\infty} p_k = 0$ and $\sum_{k=0}^{\infty} p_k = +\infty$. Hence, it can be assumed without loss of generality that $p_k \in (0,1)$ for all k.
Next, since the sequences $\{\lambda_k\}$ and $\{\alpha_k\}$ converge, we have

$$\lim_{k\to\infty}\frac{\lambda_k}{\lambda_{k+1}} = 1 \quad\text{and}\quad \lim_{k\to\infty}\frac{\alpha_k}{\alpha_{k+1}} = 1. \tag{39}$$

Combine (39) and the condition that

$$\lim_{k\to\infty}\frac{|\alpha_k - \alpha_{k+1}|}{\lambda_k\alpha_k^2} = 0,$$

we have

$$\frac{M^2|\alpha_k - \alpha_{k+1}|^2}{\alpha_k^2}.\frac{1}{p_k} = \frac{4M^2}{\gamma\tau_k}.\left(\frac{|\alpha_k - \alpha_{k+1}|}{\lambda_k\alpha_k^2}\right)^2.\frac{\lambda_k}{\lambda_{k+1}}.\frac{\alpha_k}{\alpha_{k+1}}.\lambda_k\alpha_k \to 0 \text{ as } k\to\infty, \tag{40}$$

and

$$\frac{4M^2|\alpha_k - \alpha_{k+1}|^2}{\gamma\tau_k\lambda_{k+1}\alpha_{k+1}^3}.\frac{1}{p_k} = \frac{16M^2}{\gamma^2\tau_k^2}.\left(\frac{|\alpha_k - \alpha_{k+1}|}{\lambda_k\alpha_k^2}\right)^2.\frac{\lambda_k^2}{\lambda_{k+1}^2}.\frac{\alpha_k^4}{\alpha_{k+1}^4} \to 0 \text{ as } k\to\infty. \tag{41}$$

The relations (40) and (41) together with the boundedness of $\{\tau_k\}$ infer that

$$\lim_{k\to\infty}\frac{q_k}{p_k} = 0.$$

We once again apply lemma 2 from [1] to see that $\psi_k \to 0$ as $k\to\infty$ or equivalently, $||x^k - x_{\alpha_k}||$ converges strongly to 0. The proof is complete.

References

[1] Hieu, D.V., Quy, P.K., Duong, H.N.: Equilibrium programming and new iterative methods in Hilbert spaces. Acta Applicandae Mathematicae **176**(7) (2021)